

## SKILLS

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- Robotics & Controls: Motion Planning, Path Planning, Trajectory Optimization, MPC/NMPC, LQR, PID, L1 Adaptive Control, CBF, CLF, Reinforcement Learning, Multi-Robot Coordination, Autonomous Navigation, SLAM.
- Machine Learning: TensorFlow, PyTorch, RBF Neural Networks, LSTM, Transformers, learning-based dynamics modeling, trajectory generation, data-driven control.
- Software & Tools: ROS/ROS2, C++, Python, MATLAB, Simulink, Gazebo, RViz2, CrazySwarm/CrazySwarm2, SLAM Toolbox, IPOPT, fmincon.
- Mechanical Design & Simulation: SolidWorks, Siemens NX, Teamcenter, Autodesk Inventor, ANSYS, Creo.

## EDUCATION

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**Volgograd State University, Volgograd (Russia),**  
Master of engineering  
Final thesis: Development of diesel fuel system  
Activities: Student council, Local charity organization

**GPA: 3.96**

Fall 2010-Summer 2015

**The University of Iowa, Iowa City (USA),**  
PhD Candidate, Department of Mechanical Engineering (Cooperative Autonomous Systems Lab)

**GPA: 3.95**

Winter 2023 - present

## EXPERIENCE

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**MERL, Cambridge,** Robotics Engineer Intern

January 2025 - April 2025

Implemented, integrated, and experimentally validated multi-agent trajectory planning and tracking algorithms for coordinated Crazyflie drone inspection missions in narrow and constrained environments.

- Set up and maintained the ROS2/CrazySwarm2 experimental stack for multi-Crazyflie communication, control, and flight coordination.
- Integrated research algorithms into the real-drone experimental pipeline, including trajectory execution, tracking, visualization, and testing.
- Configured SLAM Toolbox for independent drone localization and generated multiple 2D maps later merged into 3D point-cloud representations.
- Conducted RViz2-based and hardware experiments to evaluate algorithm reliability, robustness, and flight performance.
- Contributed to experimental validation for research on “Planning Multi-Agent Inspection Tours with Energy Constraints and Separation Requirements.”. The paper is submitted to IEEE Robotics and Automation

**The University of Iowa, Iowa City,** Research Assistant Robotics Lab

Winter 2023 - present

Research in robotics, autonomous systems, adaptive control, trajectory optimization, and learning-based control for AUVs and UAVs.

- Developed MPC/NMPC, LQR, PID, and L1 adaptive control algorithms for AUVs, BlueROV2, Joubert BB2, 690 AUV, and Crazyflie UAVs.
- Designed trajectory optimization and motion planning methods using Bernstein polynomials, nonlinear programming, IPOPT/fmincon, and receding-horizon control.
- Implemented ROS2-, C++-, Python-, MATLAB-, Simulink-, and Gazebo-based pipelines for simulation, real-time control, data collection, and experimental validation.
- Applied machine learning methods, including RBF neural networks, TensorFlow/PyTorch models, LSTM/Transformer-based sequence modeling, and reinforcement learning concepts for dynamics learning and trajectory generation.

- Published research on AUV autopilot systems and preparing/submitting manuscripts on UAV adaptive control, disturbance rejection, and trajectory planning.
- Selected publications:
  - “Autopilot System for Depth and Pitch Control in AUVs: Navigating Near-Surface Waves and Disturbances.”
  - “AUV Autopilot System for Controlling Depth and Pitch: Maneuvering Through Near-Surface Waves and Environmental Disturbances.”
  - Four manuscripts under conference/journal review on UAV adaptive control, disturbance rejection, and trajectory planning.

**Original-group PJSC, Moscow,** Chief of Mechanical Engineering team Winter 2020-Winter2022

Led a 24-person mechanical engineering team responsible for R&D, mechanical design, simulation, and engineering documentation for industrial robotic systems.

- Managed R&D of Delta, articulated, Cartesian, SCARA, and polar robotic systems.
- Used Autodesk Inventor, Siemens NX, Teamcenter, and ANSYS for CAD design, structural analysis, motion analysis, dynamics, stress, and deflection calculations.

**Techprom LLC, Volzhsky,** Mechanical Design Engineer Spring 2018-Summer 2019

- R&D of robots: Cartesian/Gantry, Delta robots.
- CAD Solidworks
- Simulations/Calculations in ANSYS & Simulationx (motion, statics, dynamics, stress, deflection)

**Severstal-ropes PJSC, Volgograd,** Mechanical Design Engineer Spring 2016-Spring 2018

- R&D of robots: Delta, Articulated robots.
- CAD Solidworks, Creo
- Simulations/Calculations in Simulationx (motion, statics, dynamics, stress, deflection)

## HONORS & AWARDS

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**Diploma with honors,** *Volgograd State University*

*Summer 2015*